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RESEARCH ARTICLE

A Systematic LLM-Assisted Approach for Deriving Skills From Competencies: A Case in Business Process Management Education

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ABSTRACT Competency frameworks play a central role in aligning educational objectives with professional practice. However, systematically deriving granular, assessable skills from high-level competencies remains an unresolved methodological challenge across domains. Despite the maturity of several fields, many lack structured and reproducible processes for translating competencies into measurable skills. This article introduces a systematic and generalizable method for deriving skills from competencies using Large Language Models (LLMs) guided by explicit pedagogical heuristics. The proposed pipeline was designed to ensure syntactic compliance, cognitive granularity, traceability to source materials, and controlled generation. Its primary empirical validation was conducted in the domain of Business Process Management, a mature field that nevertheless lacks a consolidated and assessable skill framework. The method was first evaluated through an internal quality analysis and expert-based validation of the generated skills, demonstrating consistent pedagogical alignment and model convergence. It was then compared against seven baseline approaches, ranging from classical text processing techniques to direct and fine-tuned LLMs. Results indicate that the heuristic-guided pipeline mitigates common limitations such as hallucination, rigidity, and loss of cognitive control, outperforming fine-tuned models in data-scarce educational contexts. Finally, to assess cross-domain generalizability, the method was applied to Environmental Education and compared against official Brazilian curriculum standards. Semantic similarity scores (0.39–0.82) showed strong alignment for broad cognitive competencies, but revealed coverage limitations for specific disciplinary skills due to corpus constraints. These findings suggest that the proposed approach provides a robust and scalable solution for competency-to-skill engineering in educational contexts.

INDEX TERMS Skills derivation, competency-based education, large language model, business process management.

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1. INTRODUCTION

In the last decade, several emerging domains in Information Systems (IS) have evolved faster than the curricular

structures needed to support their education and professional preparation [1]. Areas such as data ethics, responsible AI, sustainable business, privacy engineering, and other interdisciplinary fields increasingly demand professionals with specific competencies and measurable skills. However, despite their societal relevance, these domains typically lack consolidated competency frameworks, reference models, and proficiency structures that can guide curriculum design and assessment [2]. As a result, educational programs often rely on subjective interpretation of broad guidelines, leading to heterogeneity in course design, inconsistencies in evaluation practices, and misalignment between what is taught and what is actually required in professional contexts [3].

This challenge is not restricted to emerging domains. Even well-established disciplines within the IS field, such as Business Process Management (BPM) [4], still lack explicit competency definitions and clearly articulated skill sets. While BPM is a mature and consolidated research domain, its education still requires structured guidance on what students must be capable of performing, at which cognitive depth, and how such abilities should be assessed [5]. Such a fact hampers not only teaching and learning, but also the design of assessment instruments, benchmarking of learning outcomes across institutions, and support for continuous curriculum improvement [6].

To address this gap, this research follows a Design Science Research (DSR) method [7], [8] to develop and evaluate a systematic approach for deriving skills from previously defined competencies. Our method, which utilizes Large Language Models (LLMs) [9] as analytical support, focuses on the transformation step that is frequently underspecified in practice: turning competency-level descriptions (which tend to be coarse or heterogeneous) into concrete, measurable skills. While direct prompting of LLMs often results in generic or hallucinatory outputs, our approach employs a mathematically calibrated pedagogical heuristic to constrain the generation process. This prevents cognitive overload and ensures instructional alignment, followed by a rigorous semantic clustering phase to eliminate redundancy. These skills can then support curriculum design, assessment construction, and learning analytics. The proposed pipeline assumes that competencies are available as inputs, either extracted from existing frameworks or synthesized previously.

While the method is articulated as a generalizable pipeline, BPM serves as the primary empirical domain used to validate and refine the artifact. The artifact's development was motivated by concrete problems in BPM education, and its evaluation was guided by criteria crucial for instructional design. These criteria include coverage, understood as the extent to which the skills represent the full competency scope, granularity, referring to the adequacy of detail, and also clarity and relevance to domain practice.

The contribution of this work is fourfold:

- 1) it introduces a replicable, LLM-assisted pipeline for deriving skills that supports curriculum engineering processes;

- 2) it provides a comprehensive set of BPM skills derived and expert-validated through this method;
- 3) it offers a comparative analysis against established Natural Language Processing (NLP) [10] and generative baselines, identifying structural limitations in current automated generation techniques; and
- 4) it demonstrates the method's generalizability through a cross-domain validation in Environmental Education, evidenced by high semantic alignment with official government standards.

Given that many IS domains currently lack explicit competency models, the value of this contribution extends beyond BPM. This approach can serve as a foundation for generalizable curriculum engineering initiatives, particularly in fast-evolving domains where conceptual stabilization and competency maturity remain open challenges.

This article is organized as follows: Section II presents the related work and positions the research gap. Section III details the DSR methodology used. Section IV describes the primary artifact, the LLM-assisted skill derivation pipeline. Section V details the experimental evaluation, including the comparative analysis against seven baselines (encompassing rule-based, semantic, and generative models) and the cross-domain validation in Environmental Education. Finally, Section VI discusses the findings, implications, and concludes the article.

II. RELATED WORK

To situate our contribution, this section analyzes the literature across two critical dimensions: (1) the documented curricular gap in BPM education [1], [5], [11], [12], and (2) the nascent use of LLMs for curriculum engineering and competency analysis [2], [3], [9], [13].

A. THE CURRICULAR GAP IN BPM EDUCATION

The primary industry-facing standard in BPM is the *Guide to the Business Process Management Common Body of Knowledge* (BPM CBOK®) from the Association of BPM Professionals International (ABPMP) [11]. The CBOK is foundational, defining the essential knowledge areas (e.g., Process Modeling, Process Analysis) that serve as the basis for professional certifications.

However, its primary purpose is market validation, not pedagogical guidance. The CBOK defines what a professional should know at a high level of abstraction (competencies), but it does not provide a granular, pedagogically-grounded framework of skills. It does not specify the concrete, action-oriented, and assessable abilities required to operationalize these competencies in a higher education curriculum.

This gap is not just observational; it is confirmed by academic literature. Recent systematic mapping studies on BPM education conclude that there are “persistent challenges” and a significant “misalignment between education and anticipated job market skills” [5]. These studies explicitly highlight an “absence of consensus in defining a standardized

BPM curriculum” [5]. This reinforces the need for a systematic method to bridge the gap between high-level competency definitions (like the CBOK) and the concrete, measurable skills required in the classroom.

B. THE UNIVERSAL CHALLENGE OF SKILL DERIVATION AND EMERGENT AI METHODS

The challenge of decomposing high-level competencies into concrete, assessable skills is not unique to BPM, it is a fundamental and persistent problem in education. In mature domains such as Healthcare Education, for instance, this has long been identified as a core challenge in assessment design [14].

Recent advances in LLMs have provided a new set of tools to address this long-standing problem. However, this application remains largely exploratory. Within the field of Labor Market Intelligence, “Skill Extraction” has been formalized as a key NLP task. Researchers are actively exploring LLMs to overcome the limitations of traditional supervised models, particularly their reliance on manual annotation [9]. This focus on technical experimentation is mirrored in the BPM domain, where exploratory work has begun to leverage LLMs for related instructional design tasks [13].

Nevertheless, a common methodological gap persists. Whether dealing with the pedagogical challenges in healthcare [14] or technical NLP experiments in workforce analytics [9], the literature often prioritizes optimizing and comparing model performance. What is currently lacking is a formalized, human-in-the-loop pipeline that ensures the process is systematic, reproducible, and amenable to rigorous expert validation, shifting the focus from a mere technical experiment to a complete research methodology.

C. METHODOLOGICAL PRECEDENTS IN AUTOMATIC QUESTION GENERATION

While direct skill derivation is under-explored, the task shares conceptual proximity with Automatic Question Generation (AQG). Both domains require extracting explicit educational objectives from instructional content, adhering to criteria of semantic clarity and cognitive alignment.

As evidenced by the body of work surveyed in the systematic review [15], the AQG literature has long investigated techniques to extract action-oriented structures from text. Common approaches include statistical keyword extraction (TF-IDF) [10], graph-based centrality measures (TextRank) [16], and syntactic dependency parsing to isolate action verbs [17]. More recently, fine-tuning Language Models has been explored [18], though challenges regarding data scarcity and hallucination persist. These established methods provide the necessary theoretical baselines to evaluate the effectiveness of new LLM-assisted pipelines for skill derivation.

D. SUMMARY AND RESEARCH GAP

Synthesizing the reviewed literature, two primary gaps emerge that motivate this research:

- 1) **The Domain Gap:** BPM education presents a critical, high-impact use case characterized by a specific methodological void. While industry frameworks (such as ABPMP) define high-level expectations, systematic reviews confirm a persistent lack of curricular consensus and an unmet need for concrete, granular skill definitions [5].
- 2) **The Methodological Gap:** The decomposition of high-level competencies into assessable skills represents a fundamental challenge [14]. Although LLMs offer a promising mechanism [13], current research lacks a formalized pipeline that outperforms traditional extraction baselines (statistical or syntactic) and avoids the pitfalls of generic generation through rigorous expert validation.

The approach proposed in this article specifically targets this methodological gap. We introduce a systematic method (addressing Gap 2) designed for generalizability, and demonstrate its efficacy through application and validation within the domain of BPM education (addressing Gap 1).

III. RESEARCH METHODOLOGY

This study follows the DSR methodology, widely used in IS to create and evaluate purposeful artifacts [7]. We adopted a six-step DSR process model to guide the research, as shown in Fig. 1 and described below [8].

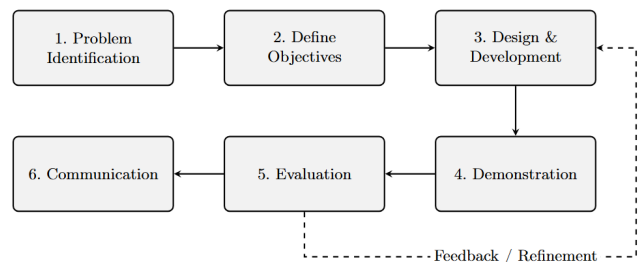


FIGURE 1. Design science research methodology process.

1. Problem Identification & Motivation:

The first step was to identify and motivate the research problem. As established in Sections II, this is characterized by a methodological gap: the lack of a systematic method to derive granular, assessable skills from high-level competency frameworks, particularly in the BPM domain.

2. Define Objectives for a Solution:

The objective was to design a method (as an artifact) that is (a) systematic and reproducible, (b) assisted by LLMs to handle complex textual analysis, (c) validated by experts, and (d) generalizable. The criteria defined in our abstract—coverage,

granularity, clarity, and relevance—served as the primary objectives for the artifact’s quality.

3. Design and Development:

This phase involved the iterative design of the artifact itself: a multi-stage pipeline. This included designing data preprocessing steps, semantic mapping strategies, heuristic modeling, and a human-in-the-loop validation workflow. The design of this artifact is the main contribution of this article and is detailed in Section IV.

4. Demonstration:

The artifact was demonstrated by applying it to the target domain of BPM education. We used the method to process a corpus of BPM textbooks and curricular guidelines to generate a concrete set of skills. This demonstration serves as a “proof-of-concept” of the method’s feasibility and utility.

5. Evaluation:

The evaluation assesses the utility and robustness of the artifact through a triangulated strategy. First, the artifact was subjected to an internal quality analysis to verify model convergence and syntactic compliance, followed by a rigorous expert-based validation to assess the intrinsic pedagogical quality of the derived skills based on instructional design criteria (relevance, clarity, and granularity). Second, to evidence the technical advantages of the proposed solution, we conducted a comparative analysis against seven baselines (encompassing rule-based, semantic, and generative models). Finally, to assess generalizability, we applied the pipeline to an external domain (Environmental Education), benchmarking the outputs against the Brazilian National Curriculum Standards [19]. This multi-faceted approach provides comprehensive evidence of the method’s efficacy, detailed in Section V.

6. Communication:

This research article constitutes the primary communication of the study. The problem definition, the artifact design (heuristic pipeline), and the rigorous experimental evaluation are detailed herein. Furthermore, to foster reproducibility and adoption, the complete source code and the generated skill datasets have been made openly available to the research community [20].

IV. THE SKILL DERIVATION METHOD: AN LLM-ASSISTED PIPELINE

This section details the design of our primary artifact: a systematic method for deriving skills from competencies. The method is structured as a pipeline (summarized in Figure 2) that combines NLP for content analysis with LLMs for skill generation, all within a human-supervised workflow. The complete process is sequentially structured into four phases

(Phases 1 through 4 in Figure 2), which are detailed in the following subsections.

A. PHASE 1: INPUT AND CONTENT HARMONIZATION

The method requires two sets of inputs: (1) a set of high-level Competencies and (2) a textual Content Corpus (e.g., textbooks, syllabi, guidelines) that describes the knowledge associated with those competencies. For the BPM application demonstrated in this article, the corpus consisted of five foundational textbooks identified through a survey of university syllabi [4], [21], [22], [23], [24].

In this phase, all textual content (e.g., chapter and section titles) is preprocessed for cross-source aggregation. This includes standard NLP techniques such as lemmatization, stopword removal, and conversion to semantic vectors using Sentence-Transformers [25].

B. PHASE 2: SEMANTIC MAPPING AND THEMATIC CLUSTERING

This phase maps the content to the competencies. First, the content is grouped into coherent Thematic Units. This is achieved by selecting a reference text (e.g., a foundational textbook) and using semantic similarity (cosine similarity) to map all other content to this reference structure. This reference structure corresponds to the thematic units derived from the reference text [4].

This process involves several steps:

- **Reference Mapping:** All content sections are mapped to the reference structure.
- **Relevance Validation:** Content with low semantic similarity to its assigned unit is discarded (we used a threshold $t_{\text{relevance}} = 0.50$).
- **Deduplication:** Agglomerative Clustering is used to merge semantically equivalent content within each unit (e.g., $t_{\text{cluster}} = 0.85$). These specific threshold values were determined empirically during pilot testing to maximize semantic cohesion while retaining the necessary volume of content for the LLM input.
- **Competency Association:** Finally, the resulting Thematic Units are semantically mapped to the high-level input Competencies.

C. PHASE 3: HEURISTIC-GUIDED SKILL GENERATION (LLM)

This phase leverages LLMs to generate the actual skill statements. To ensure consistent outputs, we adopted a heuristic-constrained approach rather than relying on unconstrained prompting or model fine-tuning. We prioritized this design because, as we empirically demonstrate in Section V, fine-tuning typically degrades performance in data-scarce domains. Consequently, to control the generation effectively, we rely on explicit constraints.

A key challenge is determining exactly how many skills (O) to generate. Generating too few leads to under-specification; generating too many induces hallucinations.

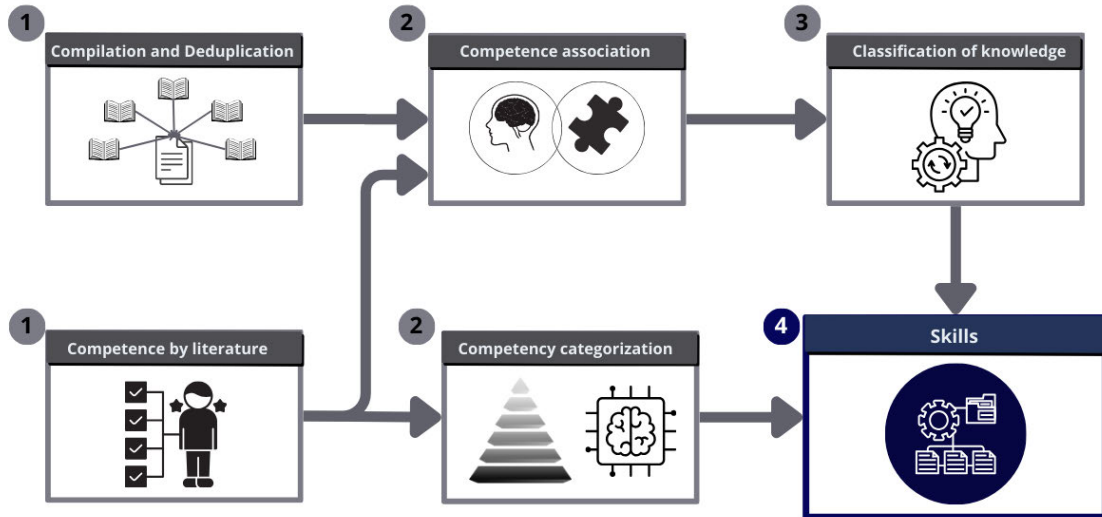


FIGURE 2. Skills generation pipeline.

To address this, we designed a pedagogical heuristic grounded in instructional design principles.

The heuristic calculates the target number of skills (O) based on four quantitative factors:

- (i) **Content Volume** (c): The count of distinct content topics (e.g., section titles, syllabus items) associated with the competency.
- (ii) **Sensitivity Factor** (λ): A calibration parameter that modulates the influence of content volume on the final output, preventing the volume metric from disproportionately dominating the skill count.
- (iii) **Cognitive Depth** (b): The target level from the Taxonomy of Educational Objectives [26].
- (iv) **Breadth** (g, d): The diversity of thematic units (g) and knowledge dimensions (d).

Formally, O is calculated as a bounded function. The outer limits (3 to 8) were empirically derived from a structural analysis of the Brazilian National Curriculum Standards. We observed that within this official framework, high-level competencies typically decompose into a minimum of 3 and an average of 8 granular skills. Crucially, this empirical range is supported by instructional design principles and cognitive load theory. Generating fewer than three skills typically indicates a competency too narrow to require decomposition, while generating more than eight introduces severe cognitive overload for the learner and makes pedagogical assessment practically unmanageable. By adopting these limits, the heuristic ensures that the output aligns with established pedagogical standards, ensuring that excessive content volume results in broader skills rather than an unmanageable list of micro-skills.

$$O = \min \left\{ 8, \max \left[3, \left\lfloor \frac{c}{\lambda} \right\rfloor + \delta_d + \delta_g + \delta_b \right] \right\} \quad (1)$$

1) PARAMETER CALIBRATION

The parameter λ acts as a divisor to scale the volume contribution. Mathematically, it functions as a necessary normalization divisor to prevent premature saturation of the generated skill count. It must be calibrated according to the average availability of topics in the dataset (C_{avg}). The goal is to select a λ that keeps the base calculation (c/λ) within the operational range (3, 8), ensuring the heuristic remains dynamically sensitive to the other cognitive and structural modifiers. We suggest the following calibration ranges:

- **Low Topic Availability** ($C_{avg} < 20$): Use $\lambda \approx 2$ or 3. For very sparse data ($C < 15$), $\lambda = 2$ allows contribution; however, as volume approaches 20, $\lambda = 3$ is preferred to prevent the base quotient from exceeding the upper bound (e.g., $20/2 = 10$, which saturates, whereas $20/3 \approx 6$, remaining within the optimal range).
- **Medium Topic Availability** ($20 \leq C_{avg} < 50$): Use $\lambda \approx 5$. This provides a balanced progression for datasets with moderate topic density.
- **High Topic Availability** ($C_{avg} \geq 50$): Use $\lambda \approx 10$. For datasets with a large number of topics (such as our corpus of detailed textbook sections), a larger divisor is necessary to prevent immediate saturation of the upper bound ($O = 8$), thereby ensuring the heuristic remains sensitive to cognitive depth modifiers.

In our experiments, the corpus comprised highly detailed BPM reference textbooks, resulting in a massive volume of extracted topics per competency. Consequently, we established the calibration factor at $\lambda = 10$. This specific divisor was mathematically necessary to normalize the dense content volume. Without it, the sheer number of extracted topics would cause the base quotient to immediately saturate at

the upper limit of 8. By applying $\lambda = 10$, we successfully prevented this premature saturation, ensuring that the final skill count generated by the heuristic remained dynamically responsive to the cognitive depth (b) and breadth (g, d) modifiers of each specific competency.

2) INTERNATIONAL GENERALIZABILITY AND RECALIBRATION

While the default clamping bounds of [3, 8] were empirically derived from the Brazilian framework (BNCC), this standard is fundamentally grounded in universal instructional design and cognitive load principles. Consequently, this range serves as a robust and globally applicable baseline for avoiding cognitive overload. Nevertheless, the heuristic remains architecturally agnostic. Researchers and practitioners applying this pipeline in different national contexts are encouraged to verify these bounds against the structural average of their local pedagogical standards and adjust the $[min, max]$ limits if necessary. Furthermore, because the sensitivity parameter λ is independent of regional context, it can be seamlessly recalibrated to any local reference corpus based purely on its topic availability (C_{avg}), following the regimes outlined above.

The modifiers δ account for specific complexities:

$$\delta_d = \begin{cases} 1, & \text{if } d > 1 \\ 0, & \text{otherwise} \end{cases},$$

$$\delta_g = \begin{cases} 1, & \text{if } g > 1 \\ 0, & \text{otherwise} \end{cases},$$

$$\delta_b = \begin{cases} -1, & \text{if } b \leq 2 \text{ (Remember or Understand)} \\ 0, & \text{if } b = 3 \text{ (Apply)} \\ 1, & \text{if } b = 4 \text{ (Analyze)} \\ 2, & \text{if } b \geq 5 \text{ (Evaluate or higher)} \end{cases}$$

Once O is calculated, a constrained prompt is engineered, instructing the LLM to generate exactly O skills. This constraint forces the model to prioritize the most critical aspects of the content while respecting the required cognitive depth defined by δ_b . The prompt structure, originally written in Portuguese and later translated into English, detailed in Table 1, enforces strict rules to prevent hallucinations and ensure syntactic uniformity.

The final prompt template was established through an iterative prompt engineering process. Exploratory testing revealed that LLM outputs were highly sensitive to the absence of specific structural constraints. For instance, without explicit instructions on verb placement and cognitive complexity (Rule 2), the models frequently buried action verbs in the middle of complex sentences or selected verbs inconsistent with the required Bloom level, thereby obscuring the intended cognitive process. Furthermore, without rigid formatting boundaries (Rule 3), the models exhibited strong conversational bias, generating unsolicited explanations and

TABLE 1. Heuristic-based prompt template used for skill generation.

Role: You are an expert in instructional design and Bloom's Taxonomy.
Target Competency: "{competency}"
Cognitive Complexity (Bloom): "{bloom_level}"
Reference Content: {content_list}
Main Task: Analyze the "Reference Content". Using only content with a direct connection to the "Target Competency", create exactly {max_skills} intermediate skills that a student needs to develop.
Rules: 1. Relevance Filter: If content does not belong to the competency, do not use it . 2. Action & Complexity: Start each skill with an action verb (infinitive) and respect the max Bloom level. 3. Output Format: List only the skills, one per line.

conversational text that significantly hindered automated extraction. Enforcing these strict rules was necessary to minimize prompt sensitivity and ensure standardized outputs.

D. PHASE 4: CONSOLIDATION AND VALIDATION

The skills generated by the LLM are provisional. This final phase consolidates and validates them to ensure the output is ready for curricular implementation.

- **Automated Consolidation (Ensemble Strategy):** To consolidate diverse outputs and mitigate single-model bias, we employed a rigorous three-step semantic clustering approach. First, for *Embedding*, skills were mapped into dense vector spaces using the paraphrase-multilingual-MiniLM-L12-v2 model [25]. Second, for *Clustering*, we applied Agglomerative Clustering (average linkage, cosine distance) with a dynamically calibrated threshold. To prevent over-compression, this threshold iterated from 0.58 to 0.70 (in 0.02 steps) until the total clusters met or exceeded the maximum skills generated by any single LLM. Finally, for *Representative Selection*, the canonical skill was extracted as the cluster medoid, the sentence demonstrating the highest average intra-cluster cosine similarity.
- **Human-in-the-Loop Validation:** The canonical skills serve as a "draft" for Subject Matter Experts (SMEs). This review process is strictly guided by the quality dimensions defined in the evaluation framework: **Relevance** (traceability to source), **Clarity** (syntactic compliance), and **Granularity** (pedagogical depth). This step is a structural feature of the method, ensuring that the final output is an expert-validated artifact rather than a raw automated suggestion.

The complete implementation of the pipeline, including the full dataset of prompts and raw response logs from the multiple LLMs employed, is openly available for reproducibility [20]. The following section details the application of this method to the BPM domain and presents a triangulated evaluation covering expert quality assessment, baseline comparison, and cross-domain generalization.

V. EXPERIMENTAL RESULTS AND EVALUATION

Following the DSR methodology (Section III), we demonstrated and evaluated the skill derivation pipeline (the artifact from Section IV) by applying it to the domain of BPM education.

A. APPLICATION SETUP: DEFINING THE INPUTS

The pipeline method (Section IV) requires two sets of inputs: (1) a set of high-level Competencies and (2) a textual Content Corpus.

To ensure rigor and address the absence of a single, universally accepted competency framework in the BPM literature, the Competencies were derived through a systematic triangulation of three main reference sources. This qualitative content analysis [12] synthesized the competencies identified in: (a) a recent systematic literature review [27], (b) the Brazilian National Curriculum Standards for computing [28], and (c) the international IS2020 competency framework [29]. This synthesis yielded the four core technical competencies that served as the primary input for our method: Model Processes, Analyze Processes, Design Processes, and Implement Processes. This set of sources is domain-dependent. In a different region (e.g., Europe), these would be substituted by equivalent national or regional competency frameworks.

The Content Corpus was constructed exclusively from structural metadata—namely, the hierarchical table of contents (chapters, sections, and subsections)—rather than from the full body text of the materials. The corpus was derived from the five BPM textbooks most frequently cited in Brazilian universities, identified through a systematic survey of course syllabi from institutions selected according to the World University Rankings (CWUR) [4], [21], [22], [23], [24], [30]. This selection is specific to the case study presented in this work. In other contexts, this component can be replaced by any structured knowledge source aligned with the target competencies, such as technical documentation or alternative bibliographic sources, thus ensuring the method's applicability beyond university syllabi.

B. RESULTS: THE DERIVED SKILL SET

The skill derivation pipeline described in Section IV was executed on October 2, 2025, using the official web interfaces of six distinct frontier LLMs: Gemini 2.5 Pro, GPT-5, DeepSeek, Qwen3-Max, Claude Sonnet 4.5, and Llama 3. Because the models were accessed via their consumer web platforms rather than specific API endpoints, the model versions correspond to the live builds active on that specific date. In accordance with the construction of the original corpus, the pipeline operated in both Portuguese and English; for clarity and consistency, all derived skills are reported in English in this article.

To promote absolute transparency and reproducibility, the complete set of interaction prompts, the direct shared links to the exact chat sessions with the LLMs, and the scripts

TABLE 2. Derived Skills: Model business processes.

Derived Skill	Qt	Cs	Traceability	Sy
Decompose complex processes into subprocesses to facilitate modeling.	5	4	"Process Decomposition" [4]	✓
Integrate events (timer, message, and error) into process modeling.	6	5	"Handling Events" / "Message Events" [4]	✓
Design processes handling events, internal, and external exceptions.	5	5	"Internal/External Exceptions" [4]	✓
Apply exclusive and inclusive gateways in process flow modeling.	4	4	"Exclusive/Inclusive Decisions" [4]	✓
Relate processes to business rules to ensure consistency.	10	4	"Processes and Business Rules" [4]	✓
Design processes with compensation mechanisms and timeouts.	1	1	"Activity Compensation" [4]	✓
Incorporate information artifacts and data objects to detail execution.	4	4	"Information Artifacts" [4]	✓
Design flexible processes considering timer events and racing events.	1	1	"Racing Events" [4]	✓
Design business processes using BPMN (Business Process Model and Notation).	9	5	"BPMN Notation" [21]	✓
Integrate process choreographies to coordinate interactions.	3	3	"Process Choreographies" [4] [21]	✓

TABLE 3. Derived Skills: Analyze business processes.

Derived Skill	Qt	Cs	Traceability	Sy
Apply root cause analysis techniques (e.g., Why-Why diagram).	7	6	"Why-Why Diagrams" / "Root Cause" [4]	✓
Analyze workflows using flow analysis and basic queuing theory.	3	3	"Flow Analysis" / "Queues" [4]	✓
Perform value analysis to distinguish value-adding activities from waste.	5	5	"Value-Added Analysis" [4]	✓
Analyze organizational process architecture based on APQC/PCF.	8	6	"Process Classification (PCF)" [23]	✓
Evaluate process performance using measures (cycle time, efficiency, BSC).	9	6	"Performance Measures" / "BSC" [4]	✓
Simulate process scenarios for behavioral analysis and bottlenecks.	2	2	"Simulation" [4]	✓
Construct case/function matrices to analyze processes.	4	4	"Case/Function Matrices" [4]	✓
Identify relevant processes for business analysis.	4	3	"Identify Processes" [4]	✓

implementing the derivation pipeline are publicly available in a GitHub repository [20].

Tables 2 through 5 present the consolidated results of the pipeline execution, organized by core competency. Each table reports the canonical formulation of the derived skills, together with indicators of cross-model convergence and theoretical grounding. After aggregation and deduplication, the final skill set comprises 35 unique skills distributed across the four core competencies.

TABLE 4. Derived Skills: Design business processes.

Derived Skill	Qt	Cs	Traceability	Sy
Apply discovery methods (interviews, workshops) to identify flows.	11	6	"Discovery Methods" [4]	✓
Create process models using methodologies and notations like UML.	11	6	"Modeling with UML" [22]	✓
Integrate process redesign heuristics to propose improvements.	12	6	"Heuristic Process Redesign" [4]	✓
Evaluate the syntactic, semantic, and pragmatic quality of models.	5	5	"Syntactic/Semantic Quality" [4]	✓
Integrate resources, activities, and events into a process architecture.	1	1	"Process Architecture" [4]	✓
Evaluate redesign needs using "The Devil's Quadrangle".	1	1	"The Devil's Quadrangle" [4]	✓
Identify the objectives and scope of the business process design project.	6	6	"Identify objectives" [22]	✓
Synthesize an improvement plan structuring necessary changes.	1	1	"Improvement plan" [22]	✓

TABLE 5. Derived Skills: Implement business processes.

Derived Skill	Qt	Cs	Traceability	Sy
Design current processes (As-Is).	2	1	"Designing current processes" [22]	✓
Apply IT tools that support process execution and management.	15	6	"IT Tools" [23]	✓
Specify business rules within implemented processes.	10	6	"Rules management" [22]	✓
Prepare a process model for execution, specifying technical properties.	5	3	"Execution Properties" [4]	✓
Identify automation boundaries in business processes.	4	4	"Automation Boundaries" [4]	✓
Use Business Process Management Systems (BPMS) for implementation.	5	5	"BPMS" [4]	✓
Monitor process performance and compliance during execution.	4	4	"Conformance/Performance" [4]	✓
Specify processes using EPC (Event-driven Process Chain).	1	1	"Processes with EPC" [22]	✓
Implement business processes according to the BPM lifecycle.	2	2	"The BPM Lifecycle" [4]	✓

The reported results are interpreted based on the following analytical elements:

- **Derived Skill:** The canonical formulation of each skill, which constitutes the primary artifact produced by the proposed method.
- **Qty (Quantity):** This column captures the frequency with which semantically equivalent formulations were generated across the models, reflecting the degree of generative redundancy observed during the pipeline execution.
- **Cs (Consensus):** This indicator expresses the level of model agreement by reporting the number of distinct

LLMs (out of six) that independently converged on the same skill, serving as a proxy for robustness.

- **Traceability (Source Topic):** This column links each derived skill to its corresponding topic in the input corpus, providing explicit evidence of forward-design alignment between the source material and the generated artifacts.
- **Sy (Syntactic Compliance):** This indicator denotes compliance (✓) with the prescribed *Action Verb* + *Object* structure, ensuring consistency with competency-based skill formulation practices.

C. INTERNAL QUALITY ANALYSIS

Before comparing the method against external baselines, the primary validation focused on the intrinsic pedagogical quality of the derived skills, establishing the evaluation criteria that are later used to analyze baseline failure modes. This evaluation was conducted by the authors, drawing on their domain expertise, using four criteria commonly adopted in competency-based instructional design: Relevance, Clarity, Coverage, and Granularity. The indicators reported in Tables 2–5 provide empirical support for this analysis by operationalizing each criterion through observable properties of the generated artifacts.

1) RELEVANCE (VIA TRACEABILITY)

Relevance was defined as alignment with established domain practice. The *Traceability* column supports this criterion by explicitly mapping each derived skill to a specific topic in the reference literature (e.g., [4], [21]). This grounding indicates that the method did not produce generic or out-of-scope hallucinations. Importantly, even highly specialized skills (e.g., "EPC Specification" or "The Devil's Quadrangle") are anchored in validated bibliographical sources, reinforcing their practical relevance.

2) CLARITY (VIA SYNTACTIC COMPLIANCE)

Clarity was defined as linguistic precision and instructional usability. The *Syn.* column shows that 100% of the derived skills conform to the instructional design structure enforced by the pipeline: *Action Verb* + *Direct Object* + *Context*. This syntactic regularity prevents the generation of vague or underspecified learning objectives (e.g., "Understand BPM"), resulting instead in skills that are actionable, unambiguous, and suitable for direct use in instructional and assessment contexts.

3) COVERAGE (VIA MODEL CONSENSUS)

Coverage was defined as the extent to which the derived skills represent the full scope of each competency. The *Cons.* column reveals a deliberate balance produced by the multi-model pipeline. Skills with high consensus (6/6 models) capture the widely recognized core of the domain (e.g., Root Cause Analysis), while low-consensus skills (1/6 models) reflect specialized or advanced technical nuances. This pattern indicates that the method captures both breadth and

depth, avoiding excessive generalization while preserving important domain-specific details.

4) GRANULARITY (VIA TOPIC MAPPING)

Granularity was defined as the adequacy of detail for instructional design and assessment construction. The *Traceability* column demonstrates that skills are mapped not only to book chapters, but to specific sub-topics (e.g., “Exclusive/Inclusive Gateways” rather than the broader category of “Modeling”). This level of specificity supports the construction of fine-grained assessment items, rubrics, and competency-based evaluation instruments, rather than remaining at overly abstract learning outcomes.

5) ADDITIONAL FINDING: REDUNDANCY (CLUSTER SIZE)

Although not a primary evaluation criterion, generative redundancy—captured by the cluster size metric (Qty)—emerges as an additional quality signal. High Qty values (e.g., 10–15) indicate that the source corpus contains dense, reinforcing language around central concepts, which contributes to the stability and robustness of the generated skills. The systematic clustering of this redundancy further demonstrates the method’s capacity to manage the large and varied outputs typical of LLM-assisted generation in a controlled, interpretable, and pedagogically meaningful manner.

Taken together, these results establish an internally validated quality baseline against which alternative approaches can be systematically contrasted, particularly with respect to rigidity, syntactic fragility, and pedagogical misalignment—dimensions that are explicitly examined in the comparative evaluation against baseline methods presented next.

D. INDEPENDENT EXPERT PANEL VALIDATION

To complement the internal quality analysis, address potential author bias, and ensure rigorous external validation, an evaluation was conducted by an independent panel of nine subject matter experts.

1) ETHICAL APPROVAL AND SURVEY DESIGN

This evaluation phase involved human participants and was formally submitted to and approved by the Research Ethics Committee of the Federal University of Rio Grande do Sul (UFRGS), under protocol CAAE 91265825.7.0000.5347. Participation was voluntary, and all respondents provided informed consent prior to initiating the survey.

The expert evaluation was conducted through a structured survey grounded in competency-based education and instructional design principles. Each generated skill was evaluated across three core dimensions: Relevance, Clarity, and Measurability, operationalized using a 5-point Likert scale. To maintain a focused and robust evaluation without causing respondent fatigue, the survey evaluated the skill set derived for Competency C1 (*Model Business Processes*), representing a central axis of the BPM domain.

2) EXPERT PANEL PROFILE

The evaluation panel consisted of nine independent subject matter experts, selected based on their active involvement in BPM education and instructional design. The methodological literature on instrument validation establishes that panels comprising five to ten experts are the gold standard to ensure analytical rigor and bias control without compromising the feasibility of data collection [31]. Table 6 details the demographic profile of the participants.

TABLE 6. Profile of the independent expert panel ($n = 9$).

Educational Role	BPM Experience	n
Corporate Instructor/Trainer	1 to 3 years	1
Corporate Instructor/Trainer	More than 10 years	1
Academic Professor	Less than 1 year	3
Academic Professor	1 to 3 years	1
Academic Professor	More than 10 years	3

This composition ensures a balanced perspective, combining the rigorous theoretical grounding of academic professors ($n = 7$) with the pragmatic insights of corporate trainers ($n = 2$). The experience distribution captures both modern educational perspectives and established pedagogical practices. The specialists’ responses are publicly available in the GitHub repository [20]

3) EXPERT VALIDATION RESULTS

To quantify the expert consensus, we calculated the Item-Level Content Validity Index (I-CVI) for each dimension. Because the evaluation employed a 5-point Likert scale, expert agreement was defined as responses of 4 (*Agree*) or 5 (*Strongly Agree*). The I-CVI for a given dimension d of a skill s is calculated as the proportion of experts in agreement:

$$I-CVI_{s,d} = \frac{n_a}{N}$$

where n_a is the number of experts who expressed agreement, and N is the total number of experts ($N = 9$).

To determine the overall pedagogical quality of each derived skill, we computed the final “Expert CVI” as the arithmetic mean of its indices across the three evaluated dimensions (Relevance, Clarity, and Measurability):

$$\text{Expert CVI}_s = \frac{I-CVI_{s,Rel} + I-CVI_{s,Clarity} + I-CVI_{s,Meas}}{3}$$

Table 7 presents the isolated CVI scores for Relevance, Clarity, and Measurability for each skill associated with C1, alongside the final calculated Expert CVI.

The results indicate a strong overall consensus among the independent experts regarding the pedagogical quality of the LLM-derived skills. Eight out of ten skills achieved an Expert CVI well above the minimum agreement threshold of 0.80 (80%), which is recommended by contemporary literature for the validation of educational artifacts and

TABLE 7. Independent expert validation (CVI Scores) for C1: Model business processes.

Skill ID	Relevance	Clarity	Measurable	Expert CVI	Status
H1_C1	1.000	1.000	0.778	0.926	Validated
H2_C1	1.000	0.889	1.000	0.963	Validated
H3_C1	0.889	0.889	0.778	0.852	Validated
H4_C1	1.000	0.875	0.889	0.921	Validated
H5_C1	1.000	0.889	0.889	0.926	Validated
H6_C1	1.000	0.778	0.667	0.815	Validated
H7_C1	0.889	0.889	0.889	0.889	Validated
H8_C1	0.889	0.667	0.556	0.704	Needs Adj.
H9_C1	1.000	1.000	1.000	1.000	Validated
H10_C1	0.889	0.667	0.556	0.704	Needs Adj.

AI-generated content [32]. This confirms their relevance, clarity, and measurability.

Skills H8_C1 and H10_C1 were flagged as requiring adjustment, primarily due to lower scores in Measurability and Clarity. This outcome is highly valuable within the DSR paradigm, as it demonstrates that while the automated pipeline generates robust baseline skills, human-in-the-loop expert oversight remains a necessary component to refine linguistic nuances and ensure the operationalization of complex pedagogical measurements.

E. COMPARATIVE EVALUATION AGAINST BASELINES

To complement the intrinsic quality analysis presented in Sections V-C and V-D, we conducted a comparative evaluation against seven baseline approaches commonly used in AQG and educational text mining. The comparison adopts the same evaluation dimensions—Relevance, Clarity, Coverage, and Granularity—thus enabling a coherent and criterion-aligned assessment.

The baselines are referenced throughout this section using the following codes:

- **B1:** Statistical keyword-based extraction (TF-IDF with handcrafted rules) [10], [16];
- **B2:** Graph-based ranking method (TextRank) [10];
- **B3:** Syntactic pattern extraction [17];
- **B4:** Hybrid symbolic–statistical approach [10], [16];
- **B5:** Vector-based semantic similarity method [24];
- **B6:** Fine-tuned Large Language Model [9], [18].
- **B7:** Direct frontier LLM prompting without heuristic constraints (Ablation Study).

Baselines B1 through B6 processed the identical input dataset (textbook corpus) under the same methodological conditions. In contrast, B7 (the ablation study) bypassed the input corpus and heuristic constraints entirely, relying solely on the models' pre-trained knowledge. To execute B7, the frontier LLMs were queried using a direct zero-shot prompt: "Act as an instructional designer. Generate the learning objectives for the BPM competency 'Model business processes', that is, the measurable skills that make up this competency." For full transparency, the complete outputs generated by all models, including the raw B7 transcripts, are available in the project's repository [20].

1) QUALITATIVE ANALYSIS OF FAILURE MODES

Tables 8, 9, and 10 present representative comparisons between baseline outputs and those generated by the proposed pipeline. These examples are not isolated cases, but rather illustrative of recurrent failure patterns observed across baseline families.

TABLE 8. Failure Analysis A: Rigidity in skill generation Input Topic: "Discovery Methods".

Method	Generated Output	Observed Defect
B4 (Hybrid)	"Summarize concepts regarding 'Discovery Methods'"	Rigid Template: Acts as a "fill-in-the-blank" exercise without pedagogical nuance.
B5 (Vector)	"Explain concepts regarding 'Discovery Methods'"	Cognitive Mismatch: Defaults to "Explain" (Bloom L2) even when the competency requires application.
Proposed	"Apply discovery methods (interviews, workshops) to identify flows."	Success: Specifies the <i>action</i> (Apply) and the <i>means</i> (interviews).

TABLE 9. Failure Analysis B: Handling nominal headings Input Topic: "APQC's Process Classification Framework".

Method	Generated Output	Observed Defect
B1 (TF-IDF)	"List the elements of 'APQC's Process Classification Framework...'"	Genericness: Merely repeats the input string without connecting it to the analysis task.
B3 (Syntactic)	[Generation Failed / Null Output]	Parser Failure: The parser could not identify a verb in the nominal heading, resulting in zero output.
Proposed	"Analyze organizational process architecture based on APQC/PCF."	Success: Infers the correct cognitive action (Analyze) from the semantic context.

TABLE 10. Failure Analysis C: Category confusion and competency drift Input Topic: "Simulating Process Scenarios / Process Modeling".

Method	Generated Output	Observed Defect
B7 (Ablation)	"Simulate a process... and recommend 3 changes that reduce cycle time by at least 15%."	Competency Drift: Drifts into the "Analyze" competency, ignoring the target "Model" constraint.
B7 (Ablation)	"Given a BPMN diagram with 5 intentional errors... propose the appropriate correction in up to 30 minutes."	Assessment Conflation: Confuses a curriculum-level skill with a highly situational test rubric, hallucinating arbitrary constraints.

The first pattern concerns template rigidity, predominantly affecting B1, B4, and B5. As illustrated in Table 8, these methods systematically generate skills following a fixed syntactic mold (e.g., "Summarize concepts regarding X"). While linguistically valid, such outputs lack instructional intent and fail to specify conditions, methods, or operational context, resulting in low pedagogical usefulness.

A second recurrent failure involves syntactic fragility, characteristic of B3. Because syntactic extractors depend

on explicit verbal predicates, they consistently fail when processing nominal headings—a dominant structure in textbooks and curricular materials (e.g., “Discovery Methods”, “Process Architecture”). As shown in Table 9, this limitation leads to null outputs or incomplete skills, directly impacting coverage.

A third critical failure pattern emerges exclusively in B7 (Table 10). While modern LLMs comprehend the domain, the absence of the pipeline’s heuristic constraints causes them to suffer from instructional category confusion, leading them to output situational assessment tasks embedded with hallucinated metrics (e.g., “reduce time by 15%”). Furthermore, without the semantic similarity routing employed by our pipeline, which strictly assigns source content to specific competencies, the unconstrained LLM suffers from *competency drift*. It frequently generates analysis or optimization skills when explicitly prompted only for modeling skills, failing to respect the boundaries of the target competency.

In contrast, the proposed method successfully infers appropriate cognitive actions from semantic context, even when the input is purely nominal, producing skills that preserve both syntactic correctness and pedagogical intent.

2) CRITERION-ALIGNED DISCUSSION OF BASELINE LIMITATIONS

Analyzing the full set of baseline outputs through the same instructional criteria used in the internal evaluation reveals profound systemic limitations across all tested approaches. These findings, which culminate in the comparative scoring matrix presented at the end of this section (Table 11), are detailed below:

a: RELEVANCE

Although most baseline methods produce outputs that are superficially related to the source content, relevance is often weak in practice. Methods B1 and B2 frequently replicate the input heading verbatim, merely attaching a generic verb (e.g., list, explain), without contextualizing the skill within domain practice. In contrast, while the direct LLM (B7) captures deep domain relevance, it introduces stochastic hallucinations, inventing arbitrary operational limits that compromise the scientific rigor of an educational curriculum. The proposed approach operationalizes relevance by explicitly mapping skills to domain actions and artifacts within the verified source corpus.

b: CLARITY

With respect to linguistic clarity, all baseline methods generate syntactically valid sentences; however, clarity in instructional design terms is frequently compromised. In particular, B1, B4, and B5 default to vague verb-object constructions, while B3 fails entirely when no verb is present. Furthermore, B7, lacking syntactic constraints, produces verbose, multi-clause paragraphs that confuse learning objectives with execution rubrics. The proposed method

consistently enforces the *Action Verb + Object + Context* structure, avoiding both null outputs and underspecified objectives.

c: COVERAGE

Baseline methods exhibit polarized coverage behavior. B3 suffers from systematic extraction failures, reducing coverage whenever nominal headings dominate. B2, conversely, produces multiple near-duplicate skills for the same topic, differing only in surface-level verb choice. This redundancy inflates apparent coverage while masking a lack of conceptual diversity. Similarly, B7 generates an overwhelming volume of highly overlapping items without structural organization. The proposed pipeline avoids these extremes by consolidating redundant generations while preserving distinct conceptual facets.

d: GRANULARITY

Granularity represents the most pronounced divergence between approaches. Baseline methods overwhelmingly operate at a coarse level, generating skills aligned with low-order cognitive processes (e.g., understand, summarize), regardless of the intended competency level. This systematic Bloom-level mismatch is evident across B1–B5. The fine-tuned LLM (B6) exacerbates this issue by producing verbose yet repetitive outputs, violating basic granularity principles such as atomicity and non-redundancy. The direct LLM (B7) equally fails in this dimension, suffering from severe granularity collapse by randomly mixing macro-strategic management competencies with software-specific micro-tasks at the exact same hierarchical level.

e: THE FINE-TUNING PARADOX

Despite expectations, B6 underperforms relative to both base models and heuristic-guided generation. Rather than learning stable instructional patterns, the fine-tuned model exhibits degeneration and overfitting, generating recursive and redundant skills. This empirically supports our design choice to favor explicit heuristic constraints (Eq. 1) over fine-tuning on limited, highly structured educational data.

The fine-tuned model corresponding to B6 was trained by the authors using competencies and skills from the BNCC as a pedagogically validated, albeit limited, reference corpus. The resulting model is publicly available on Hugging Face [33] for independent experimentation.

f: ABLATION STUDY (B7): THE NECESSITY OF CONSTRAINTS

As formally isolated by baseline B7, providing a raw frontier LLM with curriculum tasks yields encyclopedic but pedagogically unusable text. Unconstrained models inevitably suffer from instructional category confusion, metric hallucinations, and competency drift, leading to severe cognitive overload. This empirically confirms the core premise of this Design Science Research: explicit heuristic constraints (Eq. 1) and semantic consolidation pipelines are strictly necessary

to transform raw LLM generative capabilities into valid, scalable educational artifacts.

g: QUANTITATIVE SYNTHESIS OF BASELINE LIMITATIONS

To provide a structured summary of the qualitative limitations discussed above, Table 11 assigns an ordinal score (High, Medium, Low) to each baseline across the four evaluation criteria. This matrix highlights the fundamental dichotomy in current approaches: rule-based pipelines (B1–B5) systematically fail in instructional *Granularity* (due to naïve 1:1 mapping and Bloom-level mismatch), whereas generative models (B6–B7) suffer catastrophic failures in *Clarity* and *Coverage* due to unconstrained verbosity, structural degeneration, and hallucinations. These systematic failures underscore the necessity of our heuristic-constrained methodology.

TABLE 11. Comparative scoring synthesis of baselines vs. proposed method.

Method	Relevance	Clarity	Coverage	Granularity
B1-B2 (Extraction)	Low	Medium	Medium	Low
B3 (Syntactic)	Medium	Low	Low	Low
B4-B5 (Semantic)	Medium	Medium	Medium	Low
B6 (Fine-Tuned)	Medium	Low	Medium	Low
B7 (Direct LLM)	Medium	Low	Low	Low
Proposed Method	High	High	High	High

F. CROSS-DOMAIN GENERALIZATION: ENVIRONMENTAL EDUCATION

To assess whether the proposed method is overfitted to the BPM domain, we conducted a cross-domain validation in Environmental Education. In this experiment, the pipeline was applied to competencies C2 and C3 from the Natural Sciences area, using recommended textbooks as the reference corpus. The selected competencies are described in the BNCC as follows:

- **C2:** Analyze and use interpretations about the dynamics of Life, Earth, and the Cosmos to elaborate arguments, make predictions about the functioning and evolution of living beings and the Universe, and support and defend ethical and responsible decisions.
- **C3:** Investigate problem-situations and evaluate applications of scientific and technological knowledge and their implications in the world, using procedures and languages specific to Natural Sciences, to propose solutions that consider local, regional, and/or global demands, and communicate their discoveries and conclusions to varied audiences in diverse contexts.

Unlike the extensive literature available for the BPM domain, the reference corpus for Environmental Education was significantly smaller due to the limited availability of open-access, copyright-free textbooks. To account for this reduced data volume, the heuristic parameter λ was recalibrated to 2, demonstrating the method's adaptability to different data regimes. The resulting generated skills

were then compared against the BNCC, which serves as an authoritative external benchmark for educational competencies.

The validation relied on vector-based semantic similarity, measuring the alignment between the generated skills and the official BNCC competency statements. This approach allows for an objective comparison of pedagogical intent, independent of domain-specific terminology. To provide comprehensive evidence of generalizability, Table 12 reports the complete set of 12 skills generated for competencies C2 and C3, alongside their best semantic matches in the official BNCC standard.

As shown in Table 12, the similarity scores range from 0.39 to 0.82. The method achieved high semantic alignment (scores ≥ 0.75) primarily for core competencies related to explicit cognitive actions, such as scientific communication, argumentation, and interdisciplinary problem-solving. These results indicate that the heuristic pipeline successfully preserves general pedagogical intent when transferred to a distinct knowledge domain.

However, intermediate and lower scores (0.39–0.65) reveal important limitations. Where alignment was lower, the generated skills tended to focus on broad competencies highly dependent on sociocultural, historical, or affective contexts (e.g., developing mental constructs or analyzing historical contexts). These broader competencies possess a higher degree of interpretative diversity, making direct semantic alignment with specific BNCC statements more challenging.

Furthermore, an analysis of coverage reveals a concentration effect: the generated skills successfully capture central cognitive dimensions but leave highly specific disciplinary content (e.g., stellar evolution, physical phenomena, or specific biological processes) uncovered. Specifically, out of the 19 official BNCC skills associated with competencies C2 and C3, 11 did not find a direct semantic match among the generated outputs. Crucially, this coverage gap highlights an inherent characteristic of the method: the generated skills are strictly bound by the scope and volume of the input corpus. Because the reference material in this experiment was restricted to available open-access, copyright-free textbooks, the limited volume and depth of this source material naturally constrained the breadth of the resulting skills. The complete set of BNCC reference skills for these competencies, including the exact identification of all unmatched skills, is available in the project's repository [20].

Taken together, this comprehensive cross-domain evaluation demonstrates that while the method exhibits strong generalizability for extracting broad cognitive structures (critical analysis, communication), its output is highly dependent on the richness of the underlying text. It tends to underrepresent highly specific or contextualized procedural skills when the input data is sparse. This reinforces the necessity of using comprehensive corpora and applying expert validation as a complementary step to ensure full alignment with curricular frameworks.

TABLE 12. Semantic similarity between all generated skills and official BNCC standards.

Cp.	Generated Skill (Summary)	Best BNCC Match (Official)	Score
C3	Communicate scientific and technological implications using multiple languages, media and TDIC...	EM13CNT302 – Communicate results of analyses, research and experiments...	0.82
C2	Analyze and interpret scientific models, theories and evidence to explain the dynamics of Life...	EM13CNT201 – Analyze and discuss scientific models, theories and laws...	0.79
C3	Investigate real-world problems using scientific procedures, integrating interdisciplinary knowledge...	EM13CNT304 – Analyze and debate controversial situations involving applications...	0.77
C3	Analyze scientific data from reliable sources, evaluating their validity and implications...	EM13CNT303 – Interpret scientific dissemination texts considering data presentation...	0.65
C2	Relate the historical construction of scientific knowledge to sociocultural contexts...	EM13CNT201 – Analyze and discuss scientific theories across different historical periods...	0.62
C3	Evaluate teaching practices and competency development in Natural Sciences education.	EM13CNT304 – Analyze and debate controversial applications of scientific knowledge.	0.62
C3	Evaluate the reliability of scientific data and information.	EM13CNT303 – Interpret scientific dissemination texts to build strategies for selecting reliable sources.	0.61
C3	Critically evaluate technological applications considering historical development and sociocultural implications.	EM13CNT308 – Investigate and analyze technological systems and assess their social and environmental impacts.	0.58
C2	Distinguish evidence-based scientific interpretations from unsupported opinions about natural phenomena.	EM13CNT205 – Interpret results and make predictions recognizing uncertainty and explanatory limits.	0.57
C2	Formulate positions supported by reliable scientific data and evidence.	EM13CNT201 – Analyze and discuss scientific models and explanations.	0.53
C3	Analyze the historical and sociocultural context of scientific production.	EM13CNT305 – Investigate misuse of scientific knowledge in discriminatory practices.	0.52
C2	Develop critical thinking and mental constructs from questioning-based activities.	EM13CNT207 – Analyze vulnerabilities faced by youth and promote well-being actions.	0.39

VI. DISCUSSION AND CONCLUSION

This article addressed a persistent methodological gap in IS education: the lack of systematic, reproducible methods to bridge the distance between high-level competency frameworks and the granular, assessable skills required for curriculum design. Following a DSR approach, we proposed and evaluated an LLM-assisted pipeline guided by pedagogical heuristics.

A. DISCUSSION OF FINDINGS

The extensive experimental evaluation yielded three critical insights regarding the efficacy of Automated Skill Generation:

1) HEURISTICS VERSUS FINE-TUNING (THE DATA SCARCITY PROBLEM)

A major finding of this study is the counter-intuitive performance of the fine-tuned model (Baseline B6). While fine-tuning is often the standard solution for domain adaptation, our results (Section V-E) show that in data-scarce educational contexts, it leads to degradation and overfitting. In contrast, our heuristic-constrained approach (Phase 3 of the method) outperformed all seven baselines (encompassing direct LLMs, neural fine-tuning, and traditional NLP methods). By explicitly constraining the generation volume (O) to prevent cognitive overload and enforcing Bloom's taxonomy through prompt engineering, the proposed pipeline avoided the "hallucination loops" observed in fine-tuned models and the "template rigidity" characteristic of statistical methods.

2) GENERALIZABILITY ACROSS DOMAINS

Contrary to ad-hoc solutions often reported in the literature, the proposed method demonstrated cross-domain transferability. As evidenced in the Environmental Education experiment (Section V-F), the pipeline generated skills with semantic alignment scores ranging from 0.39 to 0.82 when compared to official government standards, without any domain-specific retraining. While showing strong alignment for broad cognitive competencies, it also transparently revealed coverage limitations for highly specific disciplinary skills due to input corpus constraints. This suggests that the artifact relies primarily on structural patterns of instructional design rather than memorized domain content, supporting its applicability beyond the BPM context.

3) GRANULARITY AND SYNTACTIC PRECISION

Regarding output quality, the dual evaluation process confirmed the high pedagogical rigor of the artifact. As evidenced by the CVI scores from the expert panel, the vast majority of the derived skills exhibited high relevance, clarity, and measurability, adhering to actionable syntactic structures. Furthermore, the expert validation proved its necessity by successfully identifying a small subset of skills that required pedagogical refinement (e.g., adjusting verbs for better measurability). This highlights the importance of integrating a Human-in-the-Loop phase. Overall, unlike manual extraction from syllabi, which often results in vague verbs (e.g., "Understand BPM"), the pipeline consistently produced precise, action-oriented skills suitable for instructional alignment.

B. IMPLICATIONS

1) FOR RESEARCH

This work challenges the prevailing assumption that increased model training is always beneficial in Generative AI applications. We provide empirical evidence that, for highly structured tasks such as curriculum engineering, explicit Human-in-the-Loop constraints can be more effective than model fine-tuning on limited datasets. Furthermore, the open-source pipeline [20] offers a replicable baseline for future research in Educational Data Mining and Learning Analytics.

2) FOR PRACTICE

For educators in BPM and other IS disciplines, this study delivers a structured set of 35 validated skills explicitly traceable to core reference textbooks. These skills can be directly employed to define learning outcomes, design assessments, and operationalize competency-based curricula, enabling instructors to move beyond abstract competency statements toward concrete instructional practice.

C. LIMITATIONS AND FUTURE WORK

We acknowledge limitations inherent to the design of the proposed method. First, concerning Input Dependence, the quality of the derived skills is intrinsically bounded by the quality of the input corpus (“Garbage In, Garbage Out”). If the source textbooks lack depth or specificity, the heuristic constraint (λ) will reflect this limitation accordingly in the generated output. Second, while the study validates the semantic and pedagogical quality of the derived skills through expert consensus, it does not yet assess their direct impact on student performance when applied in authentic classroom settings.

Future work will address this limitation by deploying the derived skill set in controlled educational environments to quantitatively measure learning outcomes. Additionally, we plan to extend the pipeline to automatically generate assessment items and evaluation rubrics directly from the derived skills, thereby closing the loop between automated curriculum design and practical student evaluation.

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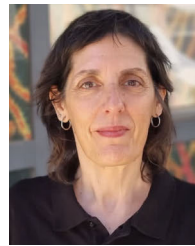
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